HDOC Day-44 - C Programming

Name: Dhruv Pandey | SAP ID: 590037660

Problem Statement

Write a menu-driven C program to perform the following operations using separate functions:

1. Reverse each word in a sentence without changing the word order.
Example: What is special? -> tahW si ?laiceps

2. Print the initials of a name.
Example: Alexander Grahambell -> A.G.

Header Files Used: stdio.h, string.h, ctype.h

C Program

```
#include <stdio.h>
#include <string.h>
#include <ctype.h>

void reverseEachWord(char sentence[])
{
    int start = 0;
    int end, i;
    char temp;

    for (i = 0; ; i++)
    {
        if (sentence[i] == ' ' || sentence[i] == '\0')
        {
            end = i - 1;

            while (start < end)
            {
                temp = sentence[start];
                sentence[start] = sentence[end];
                sentence[end] = temp;
                start++;
                end--;
            }

            start = i + 1;
        }

        if (sentence[i] == '\0')
            break;
    }
}

void printInitials(char name[])
{
    int i;

    if (name[0] != '\0')
        printf("%c.", toupper(name[0]));

    for (i = 1; name[i] != '\0'; i++)
    {
        if (name[i - 1] == ' ' && name[i] != ' ')
            printf("%c.", toupper(name[i]));
    }

    printf("\n");
}
int main()
```

```
{
    int choice;
    char sentence[200];
    char name[200];

    do
    {
        printf("\n----- MENU -----\n");
        printf("1. Reverse each word in a sentence\n");
        printf("2. Print initials of a name\n");
        printf("3. Exit\n");

        printf("Enter your choice: ");
        scanf("%d", &choice);
        getchar();

        switch (choice)
        {
            case 1:
                printf("Enter a sentence: ");
                fgets(sentence, sizeof(sentence), stdin);
                sentence[strcspn(sentence, "\n")] = '\0';

                reverseEachWord(sentence);
                printf("After reversing each word: %s\n", sentence);
                break;

            case 2:
                printf("Enter a name: ");
                fgets(name, sizeof(name), stdin);
                name[strcspn(name, "\n")] = '\0';

                printf("Initials: ");
                printInitials(name);
                break;

            case 3:
                printf("Program exited successfully.\n");
                break;

            default:
                printf("Invalid choice.\n");
        }

    } while (choice != 3);

    return 0;
}
```

Program Output

